Shameer Sheikh

Tampa, FL · Seeking Full-Time Software Engineering Roles Starting Jan. 2027 · Open To Relocation

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Education

University of South Florida — *Tampa, Florida*

Bachelor of Science, Computer Science | Minor in Psychology — *Aug. 2022 – Expected Dec. 2026*

- **GPA: 3.6/4.0 | USF Green & Gold Presidential Scholar**
- **Relevant Coursework**: Software Engineering, Data Structures and Algorithms, Generative AI, Natural Language Processing, Computer Vision, Database Design, Computer Networks, Operating Systems, Computer Architecture

Technical Skills

Programming Languages, Python, Java, C/C++, JavaScript, HTML, CSS, SQL
Frameworks & Libraries, React, Next.js, FastAPI, Spring Boot, PyTorch, NumPy, OpenCV, MediaPipe
Tools & Databases, PostgreSQL, MongoDB, Redis, Git, Docker, REST APIs, OAuth 2.0, Jupyter, Jira

Work Experience

Software Engineer Intern — *Tampa, Florida*

Ummah Professionals | JavaScript, Next.js, Supabase — *Jun. 2026 – Aug. 2026*

- Built a **three-stage recommendation system**, used by **450+ users**, for student-advisor matching: candidate generation via local **pre-scoring** (60+ advisors → top 15), **LLM**-based career-fit classification on the shortlist, and a **weighted 5-factor ranking** (career fit, industry, experience, gender preference, capacity) producing top-5 recommendations.
- Cut recommendation generation time from ∼**60s to 3–5s** and reduced **LLM API calls by 75%** by running **Gemma** (via **OpenRouter**) calls in **parallel** on a pre-scored shortlist, with results **persisted** so repeat views make no new LLM calls.
- Built the **REST API** and **Supabase** data layer, and automated monthly advisor capacity resets with **pg_cron**, removing manual updates.

Software Engineer Intern — *Tampa, Florida*

SamSearch | Python, React.js, FastAPI, PostgreSQL, OAuth 2.0, Redis/RQ — *Mar. 2026 – May. 2026*

- Shipped **HubSpot**, **Salesforce**, and **Asana** integrations for an AI-powered government contract intelligence platform used by **100+ companies**, letting teams sync pursuits and contacts to their CRM automatically through **OAuth 2.0** with **multi-tenant** token storage and **refresh.**
- Engineered a **Redis + RQ** job queue with **three dedicated workers** that moves every sync off the request path, so a slow or rate-limited provider never blocks users or other providers. Jobs fire from **six** CRUD endpoints, are gated by per-org **JSONB** toggles, and are **idempotent** (search-then-upsert), so retries and re-syncs create no duplicates.
- Delivered the **React** UI end to end (reusable connect cards, a connection-status **custom hook**), letting users connect accounts and sync records in one click from **five** existing pages.

Research Assistant — *Tampa, Florida*

Secure and Intelligent Systems Lab, University of South Florida | Python, OpenCV, NumPy, PyAV — *Jan. 2026 – Mar. 2026*

- Built **preprocessing pipelines** turning **1TB+** of maritime vessel footage (**1,000+ hrs**) into structured **face-crop datasets** using **YuNet** face detection, with **frame sampling (30x fewer frames)**, tuned **confidence thresholds**, and **fault-tolerant batch processing**, supporting crew-identification and work-hour compliance research under Dr. Yasin Yilmaz.

Research Assistant — *Tampa, Florida*

Center for Urban Transportation Research, University of South Florida — *Apr. 2026 – Present*

- Built **ground-truth datasets** at **80** field sites and **1,000+** micro-mobility devices (helmet use, direction, demographics) via synchronized video and logging, used to **train and evaluate computer-vision models**. In a follow-up **gaze study**, now annotating **eye-tracking** recordings of drivers near riders to build **labeled data** for the next models, under Dr. Changyoung Lee.

Engineering Projects

LLM Robustness Research — *Tampa, Florida*

NLP Researcher | Python, Groq API, Llama-3.3-70B, Llama-3.1-8B, Qwen3-32B, Chain-of-Thought, Self-Consistency — *Mar. 2026 - May. 2026*

- Evaluated whether **Chain-of-Thought** and **Self-Consistency** prompting recover LLM accuracy degraded by irrelevant context on the **GSM-Symbolic** benchmark, running **4,200+** inference calls across **Llama-3.3-70B, Llama-3.1-8B, and Qwen3-32B**.
- Defined a recovery-rate metric showing Self-Consistency **fully closes the accuracy gap** on **Llama-3.3-70B**; annotation of **30 CoT traces** showed models never explicitly misuse the irrelevant sentences despite the accuracy drops.

Generative AI Movie Recommendation System — *Tampa, Florida*

ML Engineer | Python, Groq/Llama-3.1-8B, FAISS, Sentence-Transformers, Pandas, NumPy — *May. 2026*

- Built a **GenAI recommendation system** that uses an **LLM** to **verbalize** each user's watch history into a **natural-language taste profile**, then **retrieves candidates** with **FAISS** and **reranks** them with the **LLM**.
- Evaluated on **MovieLens-1M** (all **6,034** users, leave-last-out), nearly doubling catalog coverage **(0.23 → 0.44)** over an **embedding-similarity** baseline while surfacing recommendations across the long tail of the **3,883**-movie catalog.

American Sign Language (ASL) Recognition System — *Tampa, Florida*

ML Engineer, Team of 4 | Python, CNN, MediaPipe, OpenCV — *Nov. 2025 - Dec. 2025*

- Developed a real-time **ASL recognition** system for **36** gestures (A–Z, 0–9) with a custom **CNN** and **MediaPipe** hand detection, running in real time on **CPU-only** hardware.
- Built an **end-to-end data processing pipeline** (hand cropping, background removal, grayscale, normalization) and trained the CNN model on **2,515 images** (72/8/20 split) with **5-frame temporal smoothing**, achieving **94% test accuracy** on unseen data.

Leadership Experience

Vice President — *Tampa, Florida*

Stampede Sounds Collective, University of South Florida — *Apr. 2024 – May. 2025*

- Served as **second-in-command** for a **14-member executive board**, overseeing strategic operations, **20+ annual activities**, and event execution for a **400+** member organization while **deputizing** for the **President** in administrative meetings.